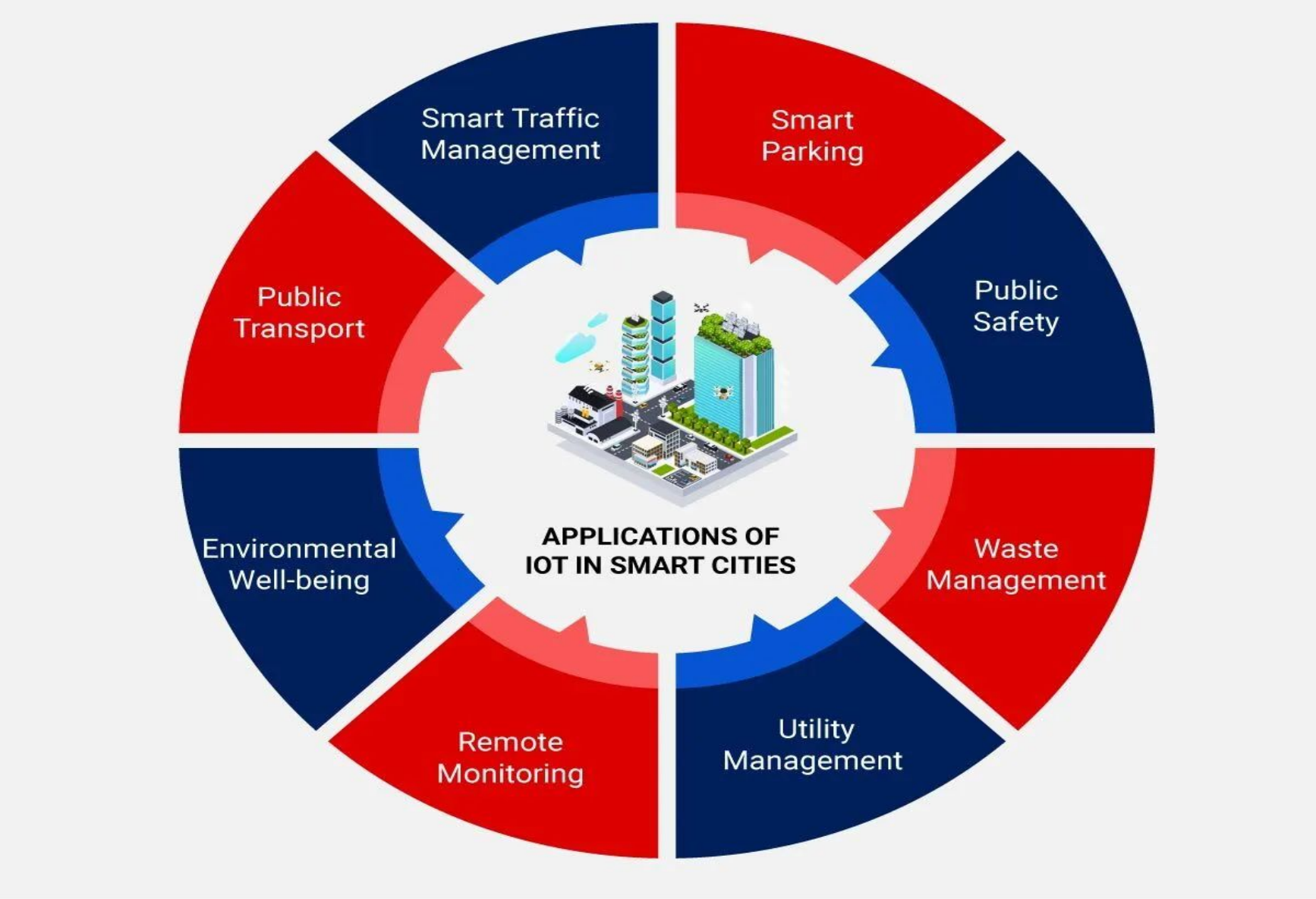
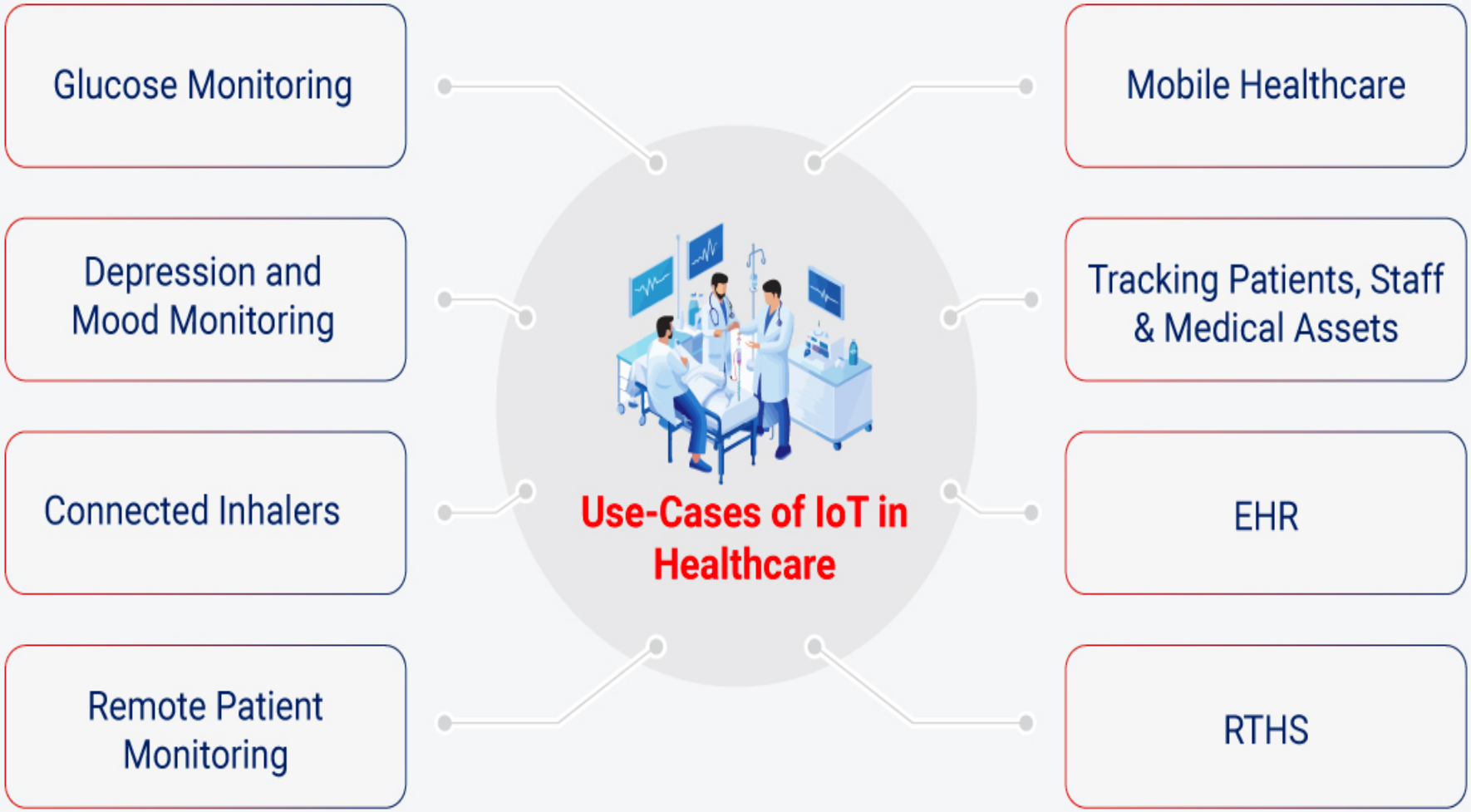
Applications of IoT

1. IoT for smart cities

- Real-world Applications of IoT in Smart Cities





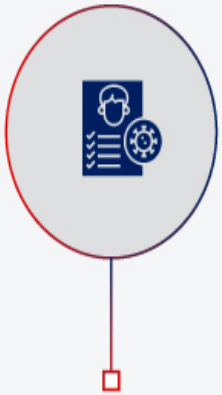
EHR: A digital, longitudinal record of a patient's health information, including demographics, medical history, allergies, medications, lab results, immunization status.

RTHS: A framework and technology paradigm that integrates data from various sources, such as Electronic Health Records (EHRs) and wearable devices, to provide continuous, actionable insights for improved patient care and operational efficiency.

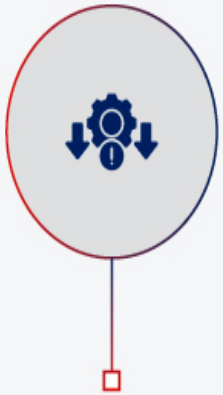
Benefits of IoT in Healthcare

Cost-efficiency

Improved Treatment



Early Disease Diagnosis



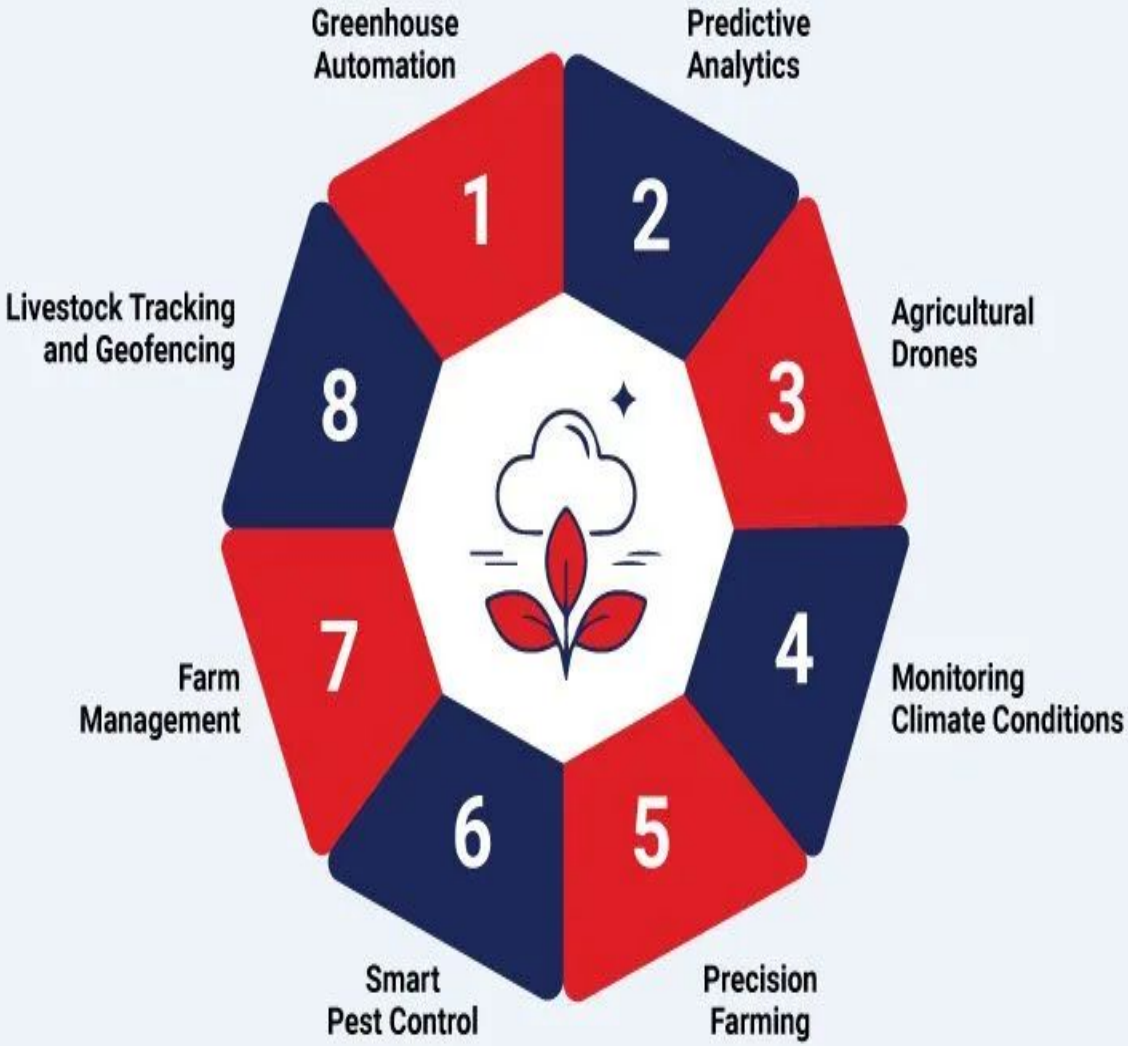
Reduction in Errors

Personalized Care

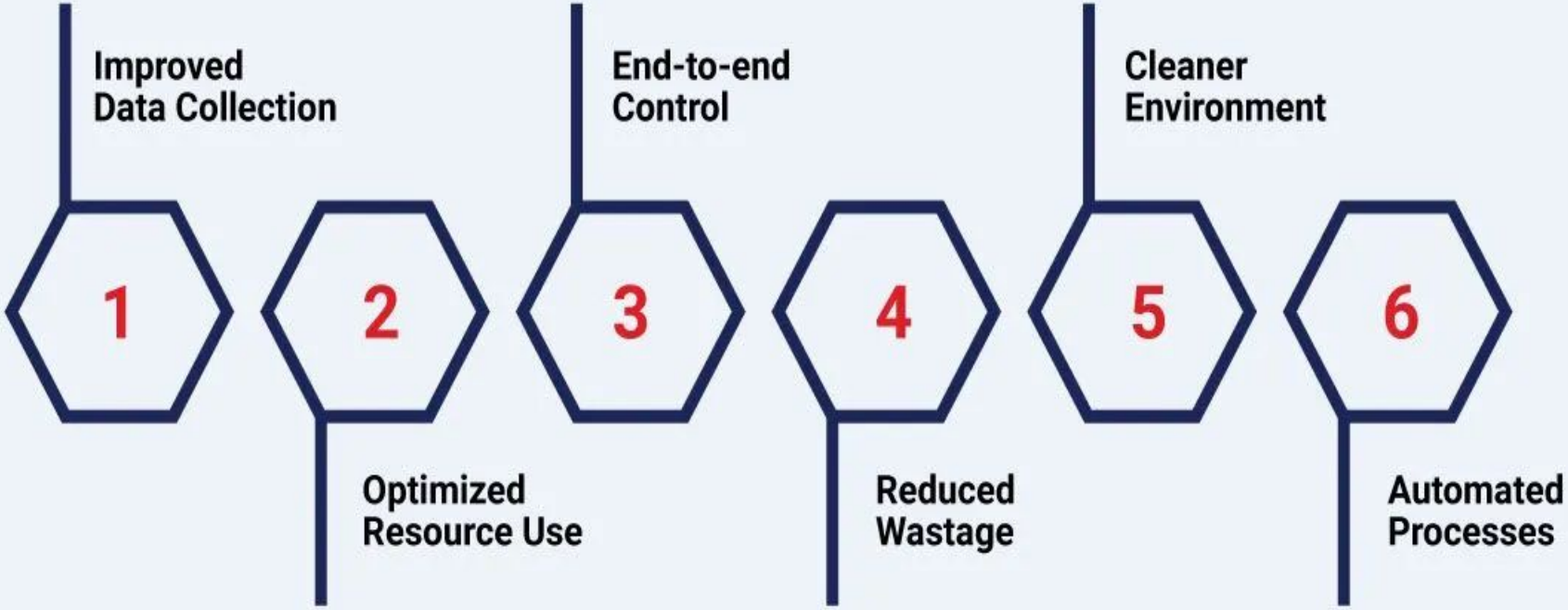
IoHT (internet of healthcare things)

- Usually, IoT products work in four stages. They are all interconnected regarding how the data is captured or processed.
- **Phase 1:** Interconnected Device Deployment: It would include sensors, actuators, monitors, detectors, and camera systems that collect the data
- **Phase 2:** Data Digitization: To aggregate & convert the digital form of analog data received from sensors & other devices
- **Phase 3:** Cloud Migration: After the received pre-processing & standardization, the data is moved to the cloud data center
- **Phase 4:** Insight Extraction: The application of advanced analytics to this final data after management & analysis helps bring about actionable insights.

Use Cases of IoT in Agriculture



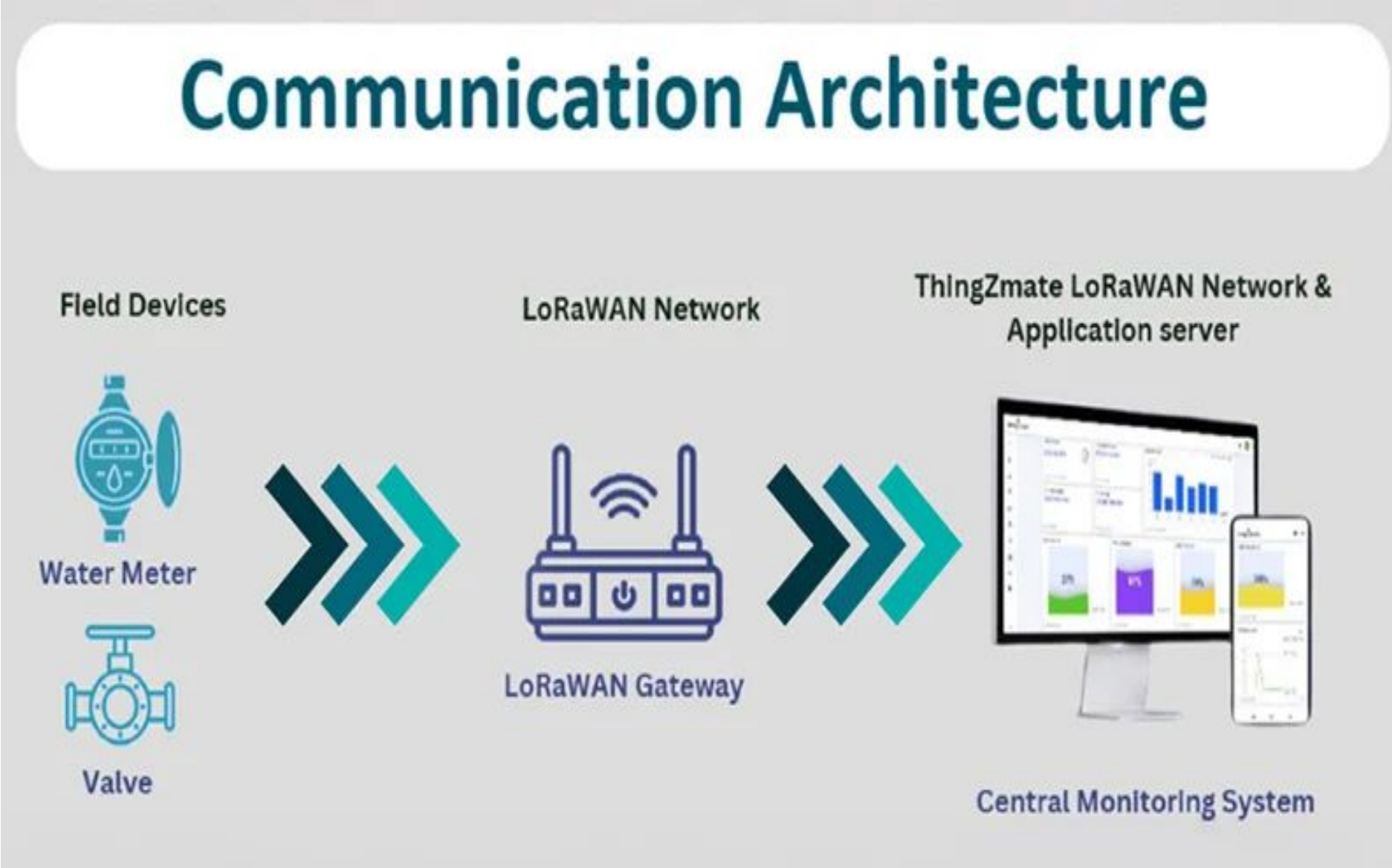
Benefits of IoT in Agriculture

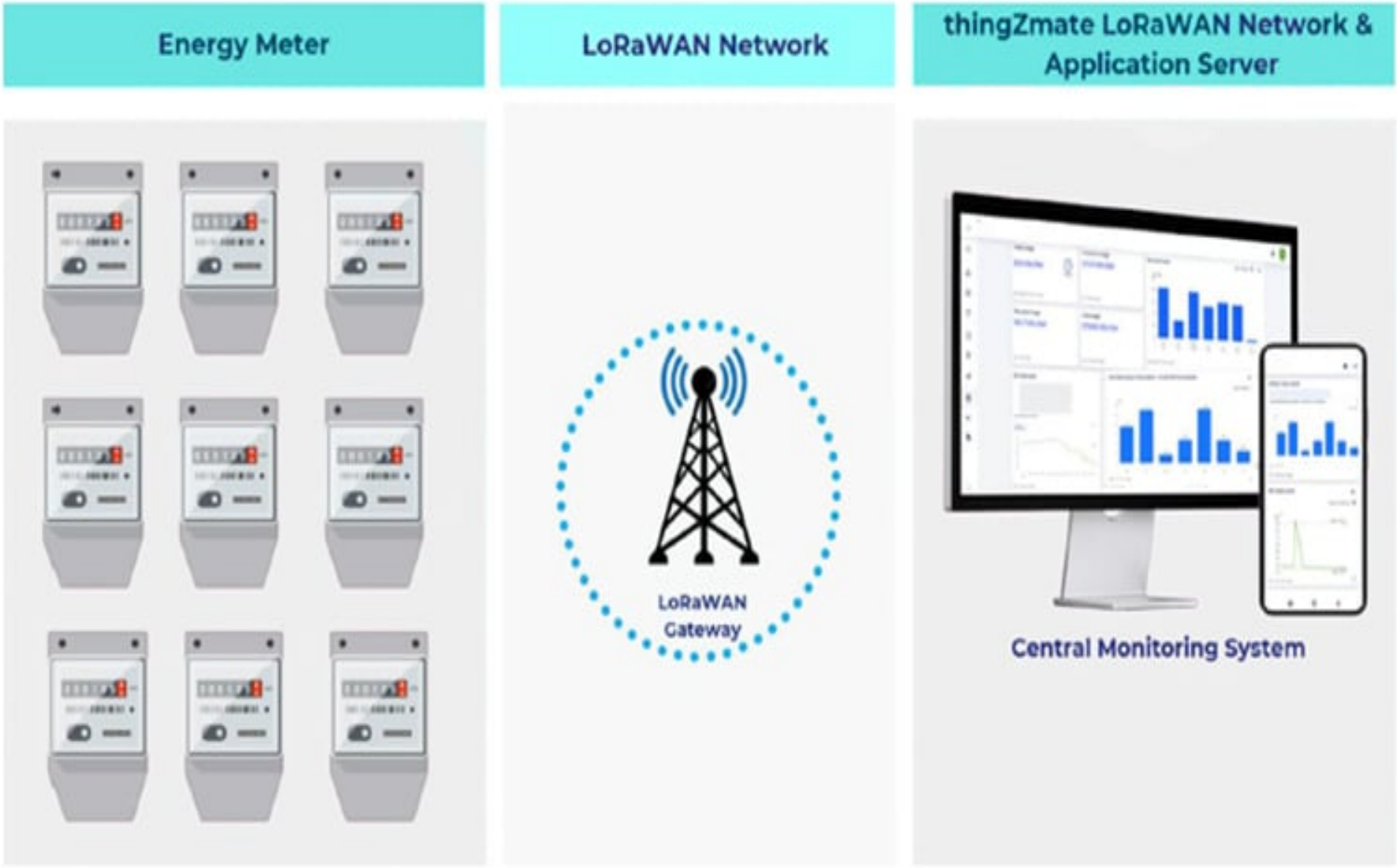


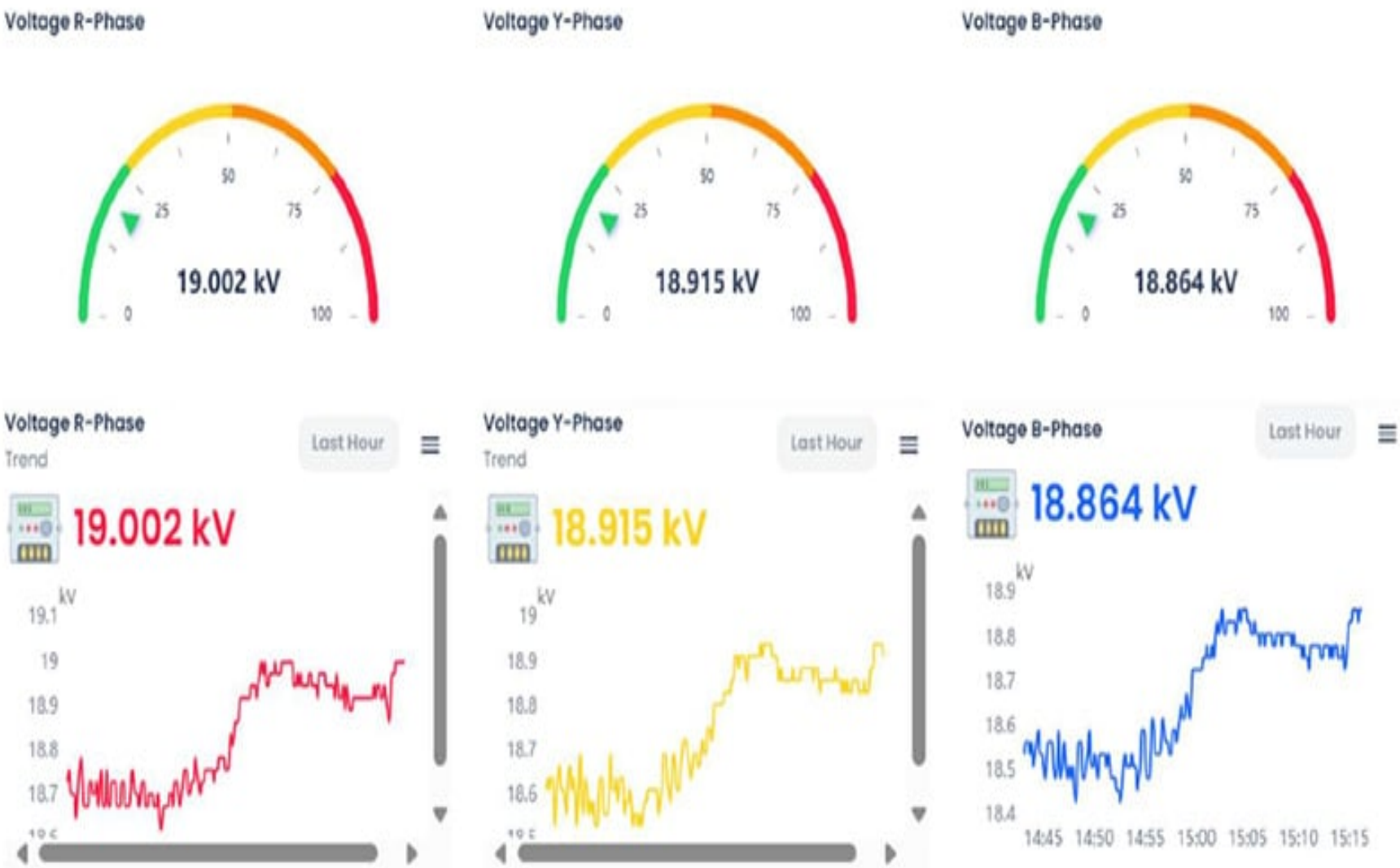
Home Automation

- IoT home automation refers to the capacity to use internet-connected equipment to operate household appliances.
- It might involve pre-programming complicated heating and lighting systems, as well as alarms and home security controls, all of which are connected via a central hub and operated remotely via a smartphone app.
- Some of the examples are: IoT home of mood-sensing music systems, smart lighting, intelligent heating and cooling, motorized blinds, and automated windows and doors etc.

Smart Metering




Energy Meter
LoRaWAN Network
thingZmate LoRaWAN Network & Application Server
LoRaWAN Gateway
Central Monitoring System



Source: https://blog.semtech.com/part-1-smart-metering-in-india-transforming-utilities-with-iot

Industrial applications of IoT

Surveillance

- Using drone for surveillance is very popular these days. But connecting IoT wit drones makes it more advance and useful in remote tracking and data sharing and storing.

Other applications…